

FIKRI AZMI NAUFAL

Data Analyst | Data Engineer | Database Administrator

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Portfolio: <https://fikriinaufal.github.io/hwJPNvOK/>

SUMMARY

Highly accomplished **Senior OSS Data Engineer & Full-Stack Developer** with extensive experience in architecting robust data solutions, automated reporting infrastructures, and enterprise data warehouses for major telecommunication projects. Recognized for expertise in **Full-Stack PHP Development**, building highly customized web-based applications and real-time monitoring dashboards tailored directly to evolving user requirements. Proven track record in orchestrating large-scale network migration data across multi-vendor environments (**Huawei, Ericsson, and ZTE**). Proficient in advanced SQL engineering, Python automation, and modern BI frameworks while pioneering the integration of **Autonomous AI Agents (CrewAI)** to streamline network analytics and drive executive decision-making.

SKILLS & COMPETENCIES

- **AI & Automation:** CrewAI (Agentic Workflows), LLM Integration, Prompt Engineering, Python Automation (Cronjobs/Schedulers)
- **Databases & Architecture:** SQL Server, PostgreSQL, MySQL, Data Warehousing, Dimensional Modeling (Star/Snowflake Schema), Fact & Dimension Tables
- **Telco OSS & NMS Platforms:** ZTE NetNumen, Nokia NetAct, ZTE U31, ZTE UME, Huawei U2000 (U2K), MML Scripting
- **Programming & Scripting:** Python (Pandas, NumPy, Matplotlib, Seaborn), PHP, Shell Scripting, SQL (Stored Procedures, Views, Triggers, Index Tuning)
- **ETL Tools:** Pentaho, Talend, SSIS (SQL Server Integration Services)
- **Data Visualization:** Power BI, Tableau, Metabase, Grafana, Apache Superset
- **Tools, Security & Platforms:** REST APIs, Role-Based Access Control (RBAC), Linux/Unix Server, Windows OS, Advanced Excel & Access
- **Languages:** Indonesian (Native), English (Professional Working Proficiency)

PROFESSIONAL WORK EXPERIENCE

Data Engineer / Analyst

June 2025 – June 2026

PT Elebram System (External Project: ZTE Indonesia)

- **Cross-Functional Data Standardization:** Collaborated closely with cross-functional network engineering teams (**RNO, NOC, and Core**) to establish strict data alignment, naming standardization, and consistency across core corporate systems during the East Java network swap.
- **Multi-Platform NMS Integration:** Integrated and centralized high-volume network data streams from legacy and new Network Management Systems, including **ZTE U31, ZTE UME, and Huawei U2000 (U2K)**, into unified MySQL repositories.

- **Performance Tuning & Query Optimization:** Engineered advanced query optimization techniques and strategic indexing, successfully accelerating data processing times and significantly improving database query performance for large-scale migration datasets.
- **Root-Cause & Alarm Correlation Analytics:** Correlated complex, historical alarm logs with real-time network performance metrics to perform deep-dive root-cause analysis on recurring network anomalies and swap-related degradations.
- **Cross-Team Optimization Support:** Partnered directly with the Radio Network Optimization (RNO) team to investigate post-swap performance issues, delivering actionable, data-driven recommendations to resolve critical network degradations.
- **Web-Based Real-Time Dashboards:** Designed, developed, and maintained robust web-based performance dashboards to provide stakeholders with real-time monitoring of critical network indicators and swap progress.
- **Automated Data Quality Governance:** Ensured high data quality and system integrity across multi-source data streams by implementing automated validation rules, duplicate filtering, and real-time anomaly detection.
- **Technical Data Governance:** Produced comprehensive technical documentation, including end-to-end ETL workflow diagrams, centralized data dictionaries, and scalable database schema designs to ensure project continuity.
- **AI Agent Development (CrewAI):** Pioneered the development and deployment of autonomous **AI Agents** utilizing **CrewAI**, leveraging advanced prompt engineering and agentic workflows to automate complex network analysis, data synthesis, and intelligent reporting tasks.
- **High-Impact Analytics Tools & Dashboards:** Designed and built a suite of proprietary network analysis tools and interactive performance dashboards that became the core standard for regional engineering teams, widely adopted by cross-functional stakeholders for high-level technical meetings and operational reviews.
- **Enterprise Backup & Disaster Recovery:** Formulated and executed robust, automated database backup strategies and disaster recovery protocols for massive multi-vendor datasets, guaranteeing 24/7 data availability, integrity, and business continuity during critical swap phases.
- **Robust Database Security Implementation:** Hardened database infrastructure by implementing advanced data security measures, including data encryption at rest/in transit, strict user access logging, and securing storage parameters against unauthorized exploits.

Data Engineer

November 2024 – May 2025

PT GMI

- **Client Liaison & Methodology Alignment:** Acted as the primary technical liaison for **Telkomsel** to align on KPI formulas, multi-vendor report templates, and complex computational methodologies across the Sulawesi, Papua, and Makassar regions.
- **Multi-Vendor Analytics (ZTE & Ericsson):** Delivered high-frequency (daily, weekly, monthly) analytical performance reports for a multi-vendor network environment, proactively executing deep-dives and root-cause analyses on network performance degradations.
- **Data Warehousing & Architecture:** Designed, constructed, and optimized scalable **MySQL data warehouses** specifically architected to handle massive volumes of analytical and business intelligence data from both ZTE and Ericsson OSS platforms.
- **Full-Stack Dashboard Development:** Developed and deployed custom, web-based KPI tracking and monitoring dashboards utilizing **PHP** and **SQL**, significantly boosting real-time operational visibility for regional engineering teams.

- **End-to-End ETL & Automation:** Built robust ETL pipelines to extract, transform, and aggregate disparate data streams, while automating recurring reporting workflows using **Python** to eliminate manual processing bottlenecks.
- **Predictive Trend Analysis:** Conducted advanced trend analysis and multi-year historical data comparisons to forecast capacity constraints and mitigate potential system or network performance issues before they impacted subscribers.

Database Engineer

May 2024 – September 2024

Meda Technology

- **Enterprise Schema Design & Optimization:** Designed and implemented optimized database schemas within **SQL Server** to efficiently store, manage, and query massive volumes of statistical data, ensuring optimal index performance and storage efficiency.
- **Data Integrity & Governance:** Enforced strict data integrity and business logic constraints by meticulously defining relational tables, data types, foreign key relationships, and data constraints.
- **Multi-Source ETL Engineering:** Constructed robust, end-to-end ETL processes to seamlessly ingest and integrate high-volume data from diverse external formats, including raw CSV files, third-party REST APIs, and legacy databases.
- **Statistical Data Transformation:** Executed rigorous data cleansing, normalization, and transformation processes to ensure high data precision, making it fully aligned with complex statistical modeling requirements.
- **Backend Database Development:** Developed and maintained highly optimized database views, complex stored procedures, and user-defined functions (UDFs) to simplify and speed up querying for downstream software applications.
- **API Integration & Data Security:** Created secure data access layers for client applications through optimized REST APIs and direct database connections while strictly enforcing **Role-Based Access Control (RBAC)** policies.
- **Disaster Recovery & High Availability:** Formulated and managed robust database backup, replication, and disaster recovery strategies to guarantee 24/7 data availability, business continuity, and minimal system downtime.

Data Engineer

November 2020 – December 2023

PT Elebram System (External Project: ZTE Indonesia)

- **Database Architecture for Network Migration:** Architected and maintained enterprise MySQL databases, managing tables, partitions, and views to guarantee high-performance data warehousing during the large-scale Nokia-to-ZTE network swap in Region Kalimantan.
- **Dimensional Data Modeling:** Structured optimized Fact and Dimension tables by applying dimensional modeling practices (Star/Snowflake schemas) to maximize BI tool query performance and streamline network migration tracking.
- **ETL Pipeline & Heavy Ingestion:** Streamlined robust ETL pipelines for large-scale data ingestion, performing heavy data aggregations on historical network alarm logs, configuration dumps, and optimization plans to support executive-level reporting.
- **Data Quality & Governance:** Conducted systematic data cleansing and rigorous data normalization routines to eliminate multi-vendor data anomalies, ensuring a single source of truth across the migration lifecycle.

- **Python & SQL Automation:** Automated end-to-end data reporting workflows via Python scripts and SQL event schedulers, removing manual bottlenecks from daily swap progress tracking.
- **Dynamic KPI Management:** Dynamically updated internal database KPI formulas and mappings in response to system version upgrades and evolving ZTE/client requirement changes during the swap project.
- **Analytical BI Dashboards:** Designed and deployed interactive analytical dashboards using **Power BI** to track migration network KPIs, significantly accelerating stakeholder and management decision-making.

Data Engineer

April 2020 – October 2020

PT Immobi Solusi Prima

- **Multi-Vendor Regional Data Integration:** Integrated heterogeneous data streams and OSS platforms from **three major telecommunication vendors (Huawei, Ericsson, and ZTE)** across multiple strategic cities under the **Telkomsel** project, ensuring cross-system data consistency.
- **Advanced Data Engineering & Analytics:** Developed and optimized complex SQL queries, views, and stored procedures to facilitate large-scale data processing, trend analysis, and advanced network analytics.
- **Heterogeneous Data Integration:** Built robust ETL pipelines to integrate disparate data streams from various operational systems and OSS platforms, ensuring cross-system data consistency and integrity.
- **Python-Driven Automation:** Designed and implemented automated reporting infrastructures utilizing **Python** and **SQL**, successfully eliminating manual reporting dependencies and reducing processing time.
- **BI Dashboard Development:** Deployed and maintained interactive, high-impact Business Intelligence dashboards using **Grafana**, and **Metabase** to deliver actionable insights to stakeholders.
- **Performance Tuning & Scalability:** Monitored, troubleshooted, and fine-tuned database operations and ETL workloads to guarantee system scalability, high availability, and efficient resource allocation.

OSS Engineer Reporting

February 2018 – March 2020

PT CCSI (External Project: ZTE Indonesia)

- **Network KPI Analysis & Optimization:** Monitored daily, weekly, and monthly network KPI trends, conducting rigorous data analysis to investigate performance anomalies; collaborated with network engineers and data scientists to implement data solutions for predictive analytics.
- **System Configuration & Automated Reporting:** Configured custom KPI formulas and structured automated reporting templates within **ZTE NetNumen** and **Nokia NetAct** Reporting Suite platforms based on Telkomsel's technical specifications.
- **Large-Scale Data Management:** Managed the daily, weekly, and monthly collection, conversion, and structural management of heavy raw network configuration dumps across 3 EMS platforms to support feature and capacity planning (Capmon).
- **Database & Server Automation:** Maintained internal **MySQL databases**, developed automated system processes using **cronjobs/schedulers**, and conducted strict data validation procedures to identify and eliminate inconsistencies.
- **Data Extraction & Technical Support:** Extracted actionable insights from OSS databases using **MML scripting** and custom **SQL queries** to fulfill high-frequency, ad-hoc requests from technical teams.

- **Fault Management & Performance Reporting:** Handled active/historical alarm collections using NetAct Fault Management, escalating critical issues to related departments while delivering critical reports including Swap Batch KPIs, Red Sector, and Cluster Health Indicators (CHI).

EDUCATION

Bachelor of Informatics Engineering
Universitas Muhammadiyah Jember

2013 – 2017